

**STRUCTURED QUESTIONS (70 MARKS)**

(INSTRUCTION: Please answer all 7 questions in the answer booklet provided.)

**QUESTION 1** [10 Marks]

- (a) Categorise the design elements labelled as (i) – (vi) into either computer architecture or computer organization in the Table 1.

- |                                    |                        |
|------------------------------------|------------------------|
| (i) ISA format                     | (iv) Cache memory type |
| (ii) Parallel/serial data transfer | (v) Data types         |
| (iii) Addressing Mode              | (vi) Read memory cycle |

[3 Marks]

Table 1: Element Design (issue)

|                       |                |
|-----------------------|----------------|
| Computer Architecture | (i) (iii) (v)  |
| Computer Organisation | (ii) (iv) (vi) |

- (b) List four (4) structural components of a computer and four (4) structural components of a CPU in Table 2.

[4 Marks]

Table 2: Structural components of Computer and CPU

|                                     |                                                                                      |
|-------------------------------------|--------------------------------------------------------------------------------------|
| Structural components of a Computer | (i) CPU<br>(ii) main memory<br>(iii) I/O system<br>(iv) system interconnection (BUS) |
| Structural components of a CPU      | (i) control unit<br>(ii) ALU<br>(iii) Registers<br>(iv) Internal Bus                 |

(c) List three (3) characteristics of Von Neumann model that was used as basis of developing a computer from first generation until recent technology in Table 3. [3 Marks]

Table 3: Characteristics of Von Neumann

|                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i) <i>Stored-Program Concept: Data and instructions are both stored in the same read-write memory, allowing the computer to be easily reprogrammed by changing the software rather than the hardware.</i>                                    |
| (ii) <i>Sequential Instruction Processing: Instructions are executed one after another in a linear sequence, governed by a program counter that tracks the address of the next instruction.</i>                                               |
| (iii) <i>Single Processing Path (Von Neumann Bottleneck): There is a single path (bus) between the CPU and memory, which means instructions and data cannot be accessed at the same time, often limiting the overall speed of the system.</i> |

| <b>QUESTION 2</b> [5 Marks] |                                                                                                                                                                                                   |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)                         | <p>Assume numbers are represented in 8-bits representation. Show the addition of the following in 1's complement: [4 Marks]</p> <p style="text-align: center;"><math>-6_{10} + 13_{10}</math></p> |
| (b)                         | <p>Represent the following 1's complement of 8-bit representation values in decimal: [1 Mark]</p> <p style="text-align: center;"><b>10100100</b></p>                                              |

**QUESTION 2** [5 Marks]

|     |                                                                                                                                                 |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | Assume numbers are represented in 8-bits representation. Show the addition of the following in 1's complement: [4 Marks]<br>$-6_{10} + 13_{10}$ |
| (b) | Represent the following 1's complement of 8-bit representation values in decimal: [1 Mark]<br><b>10100100</b>                                   |

a)  $6 \rightarrow 0000110$   
 $\rightarrow 1111001$   
 $\begin{array}{r} 1111001 \\ + 0000101 \\ \hline 10000110 \\ \hline 00000111 \end{array}$

b)  $01011011$   
 $= -91$

**QUESTION 3** [10 Marks]

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | Consider the 1 <sup>st</sup> version of Division Hardware Algorithm as shown in Figure 1 is used to perform a binary multiplication for <b>01101 / 00101</b> .<br><div><b>Steps:</b><br/><b>1 – Remainder: (R) = R – D</b><br/><b>2 – Test new R</b><br/>2a: If <math>\geq 0</math> then shift left Q (add 1 at LSB)<br/>2b: If <math>&lt; 0</math> then <math>R = D + R</math>, shift left Q (add 0 at LSB)<br/><b>3 – Shift D right</b><br/>All bits done?<br/>If still <math>&lt; (\text{max bit} + 1)</math>, repeat<br/>If <math>= (\text{max bit} + 1)</math>, stop</div><br>Figure 1: 1 <sup>st</sup> version of Division Hardware Algorithm |
| i)  | Perform and complete binary division after iteration 3 (Consider steps 1 and 2 already completed, step 3 is given) using the provided table in Table 4 on page 4. Create more rows accordingly to the numbers of iterations. [7 Marks]                                                                                                                                                                                                                                                                                                                                                                                                              |
| ii) | In the Table 4, circle the division results for quotient and remainder. Then, specify both in decimal. [3 Marks]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

a)  $01101 / 00101$

2  
5719  
- 10  
13

**QUESTION 3** [10 Marks]

- (a) Consider the 1<sup>st</sup> version of Division Hardware Algorithm as shown in Figure 1 is used to perform a binary multiplication for  $01101 / 00101$ .

**Steps:****1 – Remainder: (R) = R – D****2 – Test new R**2a: If  $\geq 0$  then shift left Q (add 1 at LSB)2b: If  $< 0$  then  $R = D + R$ , shift left Q (add 0 at LSB)**3 – Shift D right**

All bits done?

If still  $< (\text{max bit} + 1)$ , repeatIf  $= (\text{max bit} + 1)$ , stopFigure 1: 1<sup>st</sup> version of Division Hardware Algorithm

- i) Perform and complete binary division after iteration 3 (Consider steps 1 and 2 already completed, step 3 is given) using the provided table in Table 4 on page 4. Create more rows accordingly to the numbers of iterations.

[7 Marks]

- ii) In the Table 4, circle the division results for quotient and remainder. Then, specify both in decimal.

[3 Marks]



Table 4: For Question 3(a) – Division

| Iteration n | Steps                                                       | Quotient (Q) | Divisor (D)                        | Remainder (R)      |
|-------------|-------------------------------------------------------------|--------------|------------------------------------|--------------------|
| 0           | Initial value                                               | 00000        | 00101 00000                        | 00000 01101        |
| 1 & 2       | (Consider completed)                                        |              |                                    |                    |
| 3           | 1: $R = R - D$                                              |              |                                    | 11111 00101        |
|             | 2: $R = D + R$ ,<br>Shift left Q<br>Add 0 LSB               | 00000        |                                    | 00000 01101        |
|             | 3: Shift right D                                            |              | 00000 10100<br><i>(1111 01100)</i> |                    |
| 4           | $R = R - D$                                                 |              |                                    | <i>11 111 1001</i> |
|             | <i>R = D + R</i><br><i>Shift left Q</i><br><i>Add 0 LSB</i> | 00000        |                                    | <i>00000 01101</i> |
|             | Shift right                                                 |              | 00000 01010                        |                    |
| 5           |                                                             |              |                                    | <i>00000 00011</i> |
|             |                                                             | 00001        |                                    |                    |
|             |                                                             |              | 00000 00101<br><i>11111 11011</i>  |                    |
| 6           |                                                             |              |                                    | <i>11 111 1110</i> |
|             |                                                             | <i>00010</i> |                                    | <i>00000 00011</i> |
|             |                                                             |              | 00000 00010                        |                    |
|             |                                                             |              |                                    |                    |

*Quotient = 00010 = 2*

*Remainder : 00000 00011 = 3*

**QUESTION 4** [10 Marks]

|     |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                            |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | Represent decimal number <u>(-720)</u> in <u>IEEE754</u> single precision floating-point format by answering the following questions. Show all your working.                                                                                                                                                                                                                   |                                                                                                                                                            |
|     | i)                                                                                                                                                                                                                                                                                                                                                                             | Convert the number to binary. [1 Mark]                                                                                                                     |
|     | ii)                                                                                                                                                                                                                                                                                                                                                                            | Express the answer (i) in normalized form. [2 Marks]                                                                                                       |
|     | iii)                                                                                                                                                                                                                                                                                                                                                                           | Express the bit pattern in this floating-point format structure. Label the sign, biased exponent, and mantissa/significand/fraction bit clearly. [2 Marks] |
| (b) | Show how the following floating-point calculations are performed (where significands are truncated to 6 decimal digits). Show the results in normalized form. [5 Marks] <div data-bbox="557 880 1141 989" style="border: 1px solid black; padding: 10px; margin: 10px auto; width: fit-content;"> <math display="block">3.314 \times 10^1 + 822.7 \times 10^{-1}</math> </div> |                                                                                                                                                            |

**QUESTION 4** [10 Marks]

|      |                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)  | Represent decimal number <u><math>-720</math></u> in IEEE754 single precision floating-point format by answering the following questions. Show all your working.               |
| i)   | Convert the number to binary. [1 Mark]                                                                                                                                         |
| ii)  | Express the answer (i) in normalized form. [2 Marks]                                                                                                                           |
| iii) | Express the bit pattern in this floating-point format structure. Label the sign, biased exponent, and mantissa/significand/fraction bit clearly. [2 Marks]                     |
| (b)  | Show how the following floating-point calculations are performed (where <u>significands</u> are truncated to 6 decimal digits). Show the results in normalized form. [5 Marks] |
|      | <div style="border: 1px solid black; padding: 10px; width: fit-content; margin: 10px auto;"> <math>3.314 \times 10^1 + 822.7 \times 10^{-1}</math> </div>                      |

i)  $-720 \rightarrow -1011010000$

ii)  $-1.01101 \times 10^9$

sign  $\rightarrow -1 \rightarrow 1$

exponent  $\rightarrow 9$

$(2^8 - 1) = 255$

biased exponent  $\rightarrow 9 + 127 = 136$

10001000 | 01101000 0000 0000 0000 000

b) take smallest exponent

$$822.7 \times 10^{-1} \times 10^2 = 8.227 \times 10^1$$

$$\begin{array}{r} 3.314 \times 10^1 \\ + 8.227 \times 10^1 \\ \hline 11.541 \times 10^1 \end{array}$$

normalized  $\rightarrow 1.1541 \times 10^2$

$= 1.1541 \times 10^2$

$= 1.154100 \times 10^2$

QUESTION 5 [10 Marks]

Consider an x86 assembly program illustrated in Figure 2.

```
include Irvine32.inc

Str1 equ <'UTM',0>

.data
var1 WORD 2 DUP(0FFh),1000h,2000h
var2 BYTE 0FFh
msg BYTE str1
var3 WORD 4000h

.code
main PROC
MOV AX, var1
MOV AH, var2
MOV BX, var1 + 4
ADD BX, var1 + 6
MOV var3, BX
exit
main ENDP
END main
```

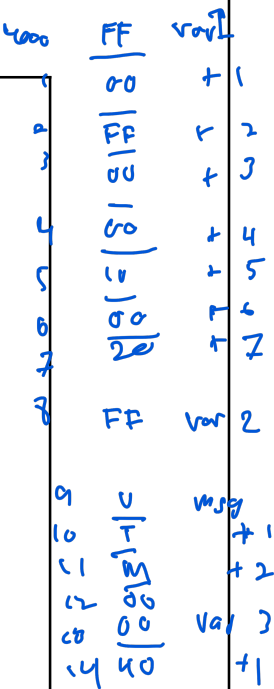


Figure 2: Assembly program

Answer the following questions based on the program.

(a) Determine the final content of the destination register in each of the instructions Table 5 after its execution. [4 Marks]

Table 5: Final content of the destination register

| Instruction |                | Register    |
|-------------|----------------|-------------|
| (i)         | MOV AX, var1   | AX = 00FF h |
| (ii)        | MOV AH, var2   | AX = FFFF h |
| (iii)       | MOV BX, var1+4 | BX = 1000 h |
| (iv)        | ADD BX, var1+6 | BX = 3000 h |

U → 41 + (21-1) = 41h + 14h = 55h

m → 41 + (13-1) = 41h + 12h = 53h

- (b) Table 6 illustrates the memory spaces horizontally starting with the memory address offset for variable var1 is 4000h. Complete the Table 6 by filling up the memory contents of all variables with the correct byte ordering. Note that ASCII code for 'A' is 41h and so on. Let the space empty for unused memory. [6 Marks]

Table 6: Memory spaces

| Offset Address | Content (Hex) |    |    |    |    |    |    |    |
|----------------|---------------|----|----|----|----|----|----|----|
| 4000h          | FF            | 0d | ef | 00 | 00 | 10 | 00 | 20 |
| 4008h          | FF            | 55 | 54 | 40 | 00 | 00 | 40 |    |

**QUESTION 6** [15 Marks]

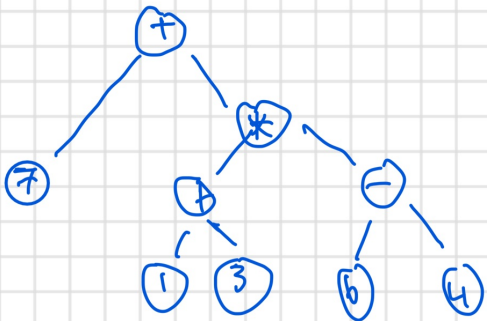
Based on the equation Y, answer the following questions:

$$Y = 7 + (1 + 3) * (6 - 4)$$

- (a) Write the Infix and construct the Expression Tree for Y. [3 Marks]
- (b) Write the Postfix and evaluate the value in Reverse Polish Notation (RPN) for Y. [2 Marks]
- (c) Convert the RPN expression from (b) into assembly codes. The instructions and registers that can be used are: PUSH, POP, IMUL (Integer Multiply), ADD, SUB, EAX and EBX. [10 Marks]

| QUESTION 6 [15 Marks]                                    |                                                                                                                                                                                            |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Based on the equation Y, answer the following questions: |                                                                                                                                                                                            |
| $Y = 7 + (1 + 3) * (6 - 4)$                              |                                                                                                                                                                                            |
| (a)                                                      | Write the Infix and construct the Expression Tree for Y. [3 Marks]                                                                                                                         |
| (b)                                                      | Write the Postfix and evaluate the value in Reverse Polish Notation (RPN) for Y. [2 Marks]                                                                                                 |
| (c)                                                      | Convert the RPN expression from (b) into assembly codes. The instructions and registers that can be used are: <u>PUSH, POP, IMUL (Integer Multiply), ADD, SUB, EAX and EBX.</u> [10 Marks] |
| push, pop → zero address                                 |                                                                                                                                                                                            |

a)  $Y = 7 + (1 + 3) * (6 - 4)$



b)  $7\ 1\ 3\ +\ 6\ 4\ -\ * +$   
 $7\ 4\ 6\ 4\ -\ * +$   
 $7\ 4\ 2\ * +$   
 $7\ 8\ +$   
 $= 15$

```

c) PUSH 7
   PUSH 1
   PUSH 3
   ADD
   PUSH 6
   PUSH 4
   SUB
   IMUL
   ADD
   POP EAX
  
```

```

c) PUSH 7
   PUSH 1
   PUSH 3
   POP EBX
   POP EAX
   ADD EAX, EBX
   PUSH EAX
   PUSH 6
   PUSH 4
   POP EBX
   POP EAX
   SUB EAX, EBX
   PUSH EAX
  
```

```

POP EBX
POP EAX
IMUL EAX, EBX
PUSH EAX
POP EBX
POP EAX
ADD EAX, EBX
PUSH EAX
  
```

QUESTION 7 [10 Marks]

Based on Figure 3, given four arrays with multiple initializers in the x86 assembly program. Assuming current pointer to the base address of the program's data segment is 0x00004000h and the address store into EAX register.

```
include Irvine32.inc

.data
count    DWORD 10h
array1   BYTE 10h, 11h, 12h, 13h
array2   WORD 123h, 234h, 345h, 456h
array3   DWORD 123456h, 23456789h

.code
main PROC
add BL, [00004005h]      ; (i)
mov ESI, offset array1   ; (ii)
mov BX, word ptr[ESI + 3] ; (iii)
mov ECX, 0
mov DX, array2[ECX]      ; (iv)
mov ECX, [EAX + 10h]     ; (v)

call dumpregs
exit
main ENDP
```

4000

10

1 00

2 00

3 00

4 10

5 11

6 12

7 13

8 23

9 01

A 34

B 02

C 45

D 03

E 56

F 04

10 56

11 34

12 12

13 00

14 89

15 67

16 45

17 28

count

+1

+2

+3

array 1

+1

+2

+3

array 2

+1

+2

+3

+4

+5

+6

+7

array 3

+1

+2

+3

+4

+5

+6

+7

Figure 3: Assembly program

Write the output for each lines labelled as (i) to (v) in Table 7.

Table 7: Output of register for each lines

| Labelled | Register | Output     |
|----------|----------|------------|
| (i)      | BL       | 11h        |
| (ii)     | ESI      | 0000 4004h |
| (iii)    | BX       | 2313h      |
| (iv)     | DX       | 0123h      |
| (v)      | ECX      | 00123456   |

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|  |
|--|
|  |
|--|

----- End of Question -----

GOOD LUCK !